

Student Social Media & Mental Health Analysis

Executive Summary

Social media has become an essential part of student life. While it provides communication, entertainment, and educational benefits, excessive usage may influence mental health, stress levels, and lifestyle habits. This project investigates the relationship between social media usage patterns and student mental well-being using a data set of 5,000 student records.

The project applies data cleaning, feature engineering, exploratory data analysis, dashboard development, and business recommendations to identify meaningful patterns and insights.

1. Business Problem

Educational institutions and mental health professionals increasingly seek to understand how digital behaviors impact student well-being. The objective of this project is to identify relationships between social media usage, sleep quality, physical activity, and mental health outcomes.

2. Project Objectives

The project aims to:

- Analyze social media usage patterns.
 - Evaluate the relationship between usage duration and mental health.
 - Identify platform-specific stress trends.
 - Examine the impact of sleep quality on mental well-being.
 - Compare mental health outcomes across student groups.
 - Develop an interactive dashboard for decision-making.
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3. Dataset Overview

The data set contains 5,000 student records and includes demographic, academic, lifestyle, and mental health variables.

Variables Included

- Age
- Gender
- Country
- Academic Level
- Most Used Platform
- Purpose of Use
- Daily Usage Hours
- Daily Unlocks
- Study Hours
- Physical Activity Hours
- Sleep Hours Per Night
- Stress Level
- Mental Health Score

4. Data Cleaning Process

The following data preparation activities were completed:

Duplicate Validation

Records were reviewed to identify duplicate entries.

Missing Value Assessment

The data set was evaluated for incomplete records and missing values.

Data Validation

Numerical fields were checked for consistency and logical ranges.

Examples:

- Age validation
- Daily Usage Hours validation
- Sleep Hours validation
- Mental Health Score validation

Standardization

Text-based fields were standardized to ensure consistency across categories.

5. Feature Engineering

To improve analysis quality, additional analytical categories were created.

Usage Category

- Low Usage (< 2 Hours)
- Medium Usage (2–5 Hours)
- High Usage (> 5 Hours)

Sleep Category

- Poor Sleep (< 6 Hours)
- Healthy Sleep ($\geq$ 6 Hours)

Mental Health Category

- Poor
- Moderate
- Good

These categories simplified trend analysis and dashboard reporting.

6. Exploratory Data Analysis

Six pivot table analyses were conducted.

Analysis 1: Mental Health by Usage Category

Objective:

Determine whether higher social media usage is associated with lower mental health scores.

Analysis 2: Mental Health by Platform

Objective:

Compare mental health outcomes across social media platforms.

Analysis 3: Stress Level by Platform

Objective:

Identify platforms associated with higher reported stress levels.

Analysis 4: Mental Health by Sleep Category

Objective:

Evaluate the relationship between sleep quality and mental well-being.

Analysis 5: Mental Health by Academic Level

Objective:

Compare mental health outcomes across academic groups.

Analysis 6: Mental Health by Gender

Objective:

Assess differences in mental health outcomes between genders.

7. Dashboard Development

An interactive dashboard was developed using Google Sheets.

Dashboard Components

- KPI Cards
- Pivot Charts
- Slicers
- Platform Analysis
- Usage Analysis
- Sleep Analysis
- Academic Level Analysis
- Gender Analysis

The dashboard allows users to explore trends and filter results dynamically.

8. Key Findings

- Students with higher social media usage generally reported lower mental health scores.
 - Healthy sleep habits were associated with better mental well-being.
 - Stress levels varied among social media platforms.
 - Academic level influenced mental health outcomes.
 - Physical activity demonstrated a positive relationship with mental health.
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9. Recommendations

Based on the analysis:

1. Encourage balanced social media usage.
2. Promote healthy sleep practices.

3. Increase awareness of mental health support resources.
 4. Encourage physical activity among students.
 5. Monitor student stress indicators and provide targeted interventions.
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10. Conclusion

This project demonstrates the complete data analytics workflow, from raw data preparation to dashboard development and business recommendations. The analysis provides valuable insights into the relationship between social media usage and student mental health and highlights the importance of healthy digital habits, sleep quality, and lifestyle behaviors in supporting student well-being.